

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Advanced Math

18

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**B**

B. $f(x) = 331.5^x$

Choice B is correct. Each of the given choices is an equation of the form $f(x) = ab^{x-k}$, where a , b , and k are constants. For an equation of this form, the coefficient, a , is equal to the value of the function when the exponent is equal to 0, or when $x = k$. It follows that in the equation $f(x) = 331.5^x$, the coefficient, 33, is equal to the value of $f(0)$. Substituting 0 for x in this equation yields $f(0) = 331.5^0$, which is equivalent to $f(0) = 331$, or $f(0) = 33$. Thus, the value of c is 33 and the equation $f(x) = 331.5^x$ shows the value of c as the coefficient.

Choice A is incorrect. This equation shows the value of $f(-1)$, not $f(0)$, as the coefficient.

Choice C is incorrect. This equation shows the value of $f(1)$, not $f(0)$, as the coefficient.

Choice D is incorrect. This equation shows the value of $f(2)$, not $f(0)$, as the coefficient.

Q2**C**

C. 1.04

Choice C is correct. A model for a quantity S that increases by a certain percentage per time period n is an exponential function in the form $S(n) = I\left(1 + \frac{r}{100}\right)^n$, where I is the initial value at time $n = 0$ for $r\%$ annual increase. It's given that the annual increase in an employee's salary is 4%, so $r = 4$. The initial value can be found by substituting 0 for n in the given function, which yields $S(0) = 38,000$. Therefore, $I = 38,000$. Substituting these values for r and I into the form of the exponential function $S(n) = I\left(1 + \frac{r}{100}\right)^n$ yields $S(n) = 38,000\left(1 + \frac{4}{100}\right)^n$, or $S(n) = 38,000(1.04)^n$. Therefore, the value of a in the given function is 1.04.

Choices A, B, and D are incorrect and may result from incorrectly representing the annual increase in the exponential function.

Q3**B****B.** $\frac{1}{4}$

Choice B is correct. It's given that the formula $P = C2^{rt}$ gives the number of bacteria in a growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. It's also given that a scientist initially measures 12,000 bacteria in the growth medium. Since the initial measurement is 0 hours after the initial measurement, it follows that when $t = 0$, $P = 12,000$. Substituting 0 for t and 12,000 for P in the given equation yields $12,000 = C2^{r0}$, or $12,000 = C2^0$, which is equivalent to $12,000 = C$. It's given that 4 hours later, the scientist measures 24,000 bacteria, or when $t = 4$, $P = 24,000$. Substituting 4 for t , 24,000 for P , and 12,000 for C in the given equation yields $24,000 = 12,0002^{4r}$. Dividing each side of this equation by 12,000 yields $2 = 2^{4r}$, or $2^1 = 2^{4r}$, which is equivalent to $1 = 4r$. Dividing both sides of this equation by 4 yields $\frac{1}{4} = r$. Therefore, the value of r is $\frac{1}{4}$.

Choice A is incorrect. This is the value of the reciprocal of C .

Choice C is incorrect. This is the value of the reciprocal of r .

Choice D is incorrect. This is the value of C .

Q4**D**

D. $p = 15,000(1.04)^x$

Choice D is correct. It's given that a model predicts the population of Bergen in 2005 was 15,000. The model also predicts that each year for the next 5 years, the population increased by 4% of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by 1.04. Since the predicted population in 2005 was 15,000, the predicted population 1 year later is $15,000(1.04)$. The predicted population 2 years later is this value times 1.04, which is $15,000(1.04)^2$. The predicted population 3 years later is this value times 1.04, or $15,000(1.04)^3$. More generally, the predicted population, p , x years after 2005 is represented by the equation $p = 15,000(1.04)^x$.

Choice A is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 0.96 rather than 15,000.

Choice B is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 1.04 rather than 15,000.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by 4% each year.

Q5**D**

D. $gx = 7x^3 - 2$

Choice D is correct. If the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down k units in the xy -plane, the function g can be defined by an equation of the form $gx = fx - k$. It's given that $fx = 7x^3$ and the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down 2 units. Substituting $7x^3$ for fx and 2 for k in the equation $gx = fx - k$ yields $gx = 7x^3 - 2$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This equation defines a function g for which the graph of $y = gx$ is the result of shifting the graph of $y = fx$ up, not down, 2 units.

Q6**-6**

Accept also: 4

The correct answer is either -6 or 4. The x -intercepts of a graph in the xy -plane are the points where $y = 0$. Thus, for an x -intercept of the graph of $y = f(x)$, $0 = f(x)$. Substituting 0 for $f(x)$ in the equation $f(x) = x + 6x - 4$ yields $0 = x + 6x - 4$. By the zero product property, $x + 6 = 0$ and $x - 4 = 0$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the x -coordinates of the x -intercepts of the graph of $y = f(x)$ are -6 and 4. Note that -6 and 4 are examples of ways to enter a correct answer.

Q7**D**

D.

x	-1	0	1
$q(x)$	16	32	64

Choice D is correct. Substituting -1 for x in the given function yields $q(-1) = 322^{-1}$, which is equivalent to $q(-1) = 32^{-1}_2$, or $q(-1) = 16$. Therefore, when $x = -1$, the corresponding value of qx for function q is 16. Substituting 0 for x in the given function yields $q(0) = 322^0$, which is equivalent to $q(0) = 32^1$, or $q(0) = 32$. Therefore, when $x = 0$, the corresponding value of qx for function q is 32. Substituting 1 for x in the given function yields $q(1) = 322^1$, which is equivalent to $q(1) = 32^2$, or $q(1) = 64$. Therefore, when $x = 1$, the corresponding value of qx for function q is 64. Of the choices given, only the table in choice D gives these three values of x and their corresponding values of qx for function q .

Choice A is incorrect. This table gives three values of x and their corresponding values of qx for the function $qx = 322x$.

Choice B is incorrect. This table gives three values of x and their corresponding values of qx for the function $qx = 232^x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8**A****A.** 3

Choice A is correct. It's given that an object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. This relationship can be represented by the equation $K = \frac{1}{2}mv^2$, where K is the kinetic energy, m is the mass, and v is the speed. Substituting a mass of 4 kilograms for m and a kinetic energy of 18 joules for K results in the equation $18 = \left(\frac{1}{2}\right)(4)v^2$, or $18 = 2v^2$. Dividing both sides of this equation by 2 yields $9 = v^2$. Taking the square root of both sides yields $v = -3$ and $v = 3$. Since speed can't be expressed as a negative number, the speed of the object is 3 meters per second.

Choice B is incorrect and may result from computation errors. Choice C is incorrect. This is the value of v^2 rather than v . Choice D is incorrect. This is the value of $4v^2$ rather than v .

Q9**C****C.** $w + 29$

Choice C is correct. It's given that the expression $ww + 29$ represents the area, in square centimeters, of a rectangular painting, where w is the width, in centimeters, of the painting. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the painting is represented by the expression $w + 29$.

Choice A is incorrect. This expression represents the width, in centimeters, of the painting, not its length, in centimeters.

Choice B is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the painting, not its length, in centimeters.

Choice D is incorrect. This expression represents the area, in square centimeters, of the painting, not its length, in centimeters.

Q10

B

B. $c = 1,3001.07^m$

Choice B is correct. It's given that in January 2018, there were 1,300 customers subscribed to a company's newsletter and for the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a1 + \frac{r}{100}^m$, where c is the total number of customers subscribed to the company's newsletter m months after January 2018, a is the number of customers subscribed to the newsletter in January 2018, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting 1,300 for a and 7 for r in this equation yields $c = 1,3001 + \frac{7}{100}^m$, or $c = 1,3001.07^m$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was 93% less, not 7% greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was 70%, not 7%, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was 600%, not 7%, greater than the total number subscribed the previous month.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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